Q.30 Explain Zeroth law of thermodynamics.

Q.31 Explain the use of steam stop value with its diagram.

Q.32 Classify the steam boilers.

Q.33 Describe working of reciprocating air compressor.

Q.34 Explain Boyle's law.

Q.35 Write shot note on heat transfer through convection.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain construction and working babcock & wilcox boiler with neat sketch.

Q.37 Define the following terms (i)wet steam (ii) dry steam (iii) superheated steam and (iv) specific volume to steam

Q.38 Explain first law of TD (Joule's Experiment) with neat diagram.

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Roll No. ....

2nd Sem / Mechanical Engineering (MSIL)
Sub : Thermodynamics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Constant volume process is called as

a) Isochoric process b) Isobaric process

c) Isothermal process d) Relative process

Q.2 Water tubes in simple vertical boiler are

a) Inclined b) Vertical

c) Tilted d) Horizontal

Q.3 For complete specification of superheated vapour, one needs (CO4)

a) Pressure

b) Temperature

c) Both pressure and temperature

d) Specific volume

Q.4 The S.I. unit of characteristic gas constant is (CO3)

a) J/kg b) kJ/kg.

c) J/kg K. d) J/K.

Q.5 A gas which obeys all the gas laws under all conditions of temperature and pressure is called

a) Perfect gas b) Natural gas

c) Real gas d) All of the above

Q.6 The triple point of water is

a) 52.25K b) 273.16K

c) 752.25K d) 96.60K

Q.7 Second law of thermodynamics defines

a) Internal energy b) Entropy

c) Temperature d) Heat

Q.8 Which of the following is a boiler mounting called?

a) Economiser b) Pressure gauge

c) Super heater d) Preheater

Q.9 Heat transfer in liquids takes place by

a) Conduction b) Convection

c) Radiation d) None of these

Q.10 Which of the following is a good conductor of heat?

a) Wood b) Rubber

c) Metal d) Non metal

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define thermal conductivity.

Q.12 Give an example of water tube boiler.

Q.13 Define Mollier diagram.

Q.14 Define constant temperature process.

Q.15 Define modern boilers.

Q.16 What is the function of fusible plug?

Q.17 What is externally fired boiler?

Q.18 Define closed thermodynamic system.

Q.19 State Charle's Law.

Q.20 Explain radiation.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What is PMM of first and second kind?

Q.22 Explain the concept of sensible heat.

Q.23 What is constant temperature process? Show its PV diagram.

Q.24 Explain second law of thermodynamics.

Q.25 Write the uses of compressed air.

Q.26 Give few examples of Accessories.

Q.27 Why accessories are used for a boiler?

Q.28 Write short note on P.V.T. surface.

Q.29 Why multistage compression is preferred over single stage compression?